Design and Development of a Web-Based Student Attendance Management System

Md. Mainul Islam (ID: 23203037)
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A Thesis in the Partial Fulfillment of the Requirements
for the Award of Bachelor of Computer Science and Engineering (BCSE)



Department of Computer Science and Engineering
College of Engineering and Technology
IUBAT - International University of Business Agriculture and Technology

Fall 2026

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The thesis has been examined and approved,

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Prof Dr. Utpal Kanti Das
Chairman & Professor
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Fall 2026

LETTER OF TRANSMITTAL

8th November, 2026

The Chairman
Thesis Defense Committee
College of Engineering and Technology (CEAT)
IUBAT - International University of Business Agriculture and Technology
4 Embankment Drive Road, Sector 10, Uttara Model Town
Dhaka 1230, Bangladesh

Subject: Submission of the thesis report.

Sir,

With due respect, we are pleased to submit our report titled **"Design and Development of a Web-Based Student Attendance Management System"** as a part of the requirements for the Bachelor of Computer Science and Engineering (BCSE) program.

Working on this topic helped us understand how a research report is planned and written. We learned how to define a problem, set clear objectives and explain our approach in simple steps. We hope the report meets your expectations.

We would be very grateful if you kindly go through the report and give us your valuable feedback.

Thank you.

Md. Mainul Islam
ID: 23203037

Jannati Ferdous
ID: 23303157

STUDENT'S DECLARATION

We, Md. Mainul Islam (ID: 23203037) and Jannati Ferdous (ID: 23303157), are students of the Bachelor of Computer Science and Engineering (BCSE) program under the College of Engineering and Technology (CEAT) of IUBAT - International University of Business Agriculture and Technology. We declare that this report, titled **"Design and Development of a Web-Based Student Attendance Management System"**, has been prepared by us as a part of our thesis course.

The work in this report is our own. Where we have used ideas or information from other sources, we have mentioned them in the references.

This report has not been submitted anywhere else for any other degree, award or presentation.

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Md. Mainul Islam
ID: 23203037

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Jannati Ferdous
ID: 23303157

ACKNOWLEDGMENTS

First of all, we thank Almighty Allah for giving us the strength and patience to complete this report.

We would like to express our sincere gratitude to our course teacher and thesis supervisor for the guidance, advice and encouragement we received throughout this work. The clear instructions on how to plan and write a thesis report made our task much easier.

We are also thankful to the Chairman and the Coordinator of the Department of Computer Science and Engineering, IUBAT, for their support. We thank all our teachers, who taught us the basics that we used in this work.

Finally, we thank our families and friends for their love, prayers and support. Without them, this work would not have been possible.

SUPERVISOR'S CERTIFICATION

This is to certify that the thesis report titled **"Design and Development of a Web-Based Student Attendance Management System"** has been prepared by Md. Mainul Islam (ID: 23203037) and Jannati Ferdous (ID: 23303157) of IUBAT - International University of Business Agriculture and Technology, as a part of the requirements of the Bachelor of Computer Science and Engineering (BCSE) program.

The report was prepared under my supervision. To the best of my knowledge and according to the declaration of the students, no part of this report has been submitted anywhere for any degree, diploma or certificate.

The students are allowed to submit the report. I wish them every success in their future work.

Thesis Supervisor

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DEPARTMENT'S CERTIFICATION

On behalf of the Department of Computer Science and Engineering, IUBAT - International University of Business Agriculture and Technology, we confirm that the thesis report **"Design and Development of a Web-Based Student Attendance Management System"** for the Bachelor of Computer Science and Engineering (BCSE) degree was duly presented by Md. Mainul Islam (ID: 23203037) and Jannati Ferdous (ID: 23303157) and approved by the department.

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ABSTRACT

Taking attendance is a daily task in every university class. In many classes it is still done on paper. This takes class time, the sheets can be lost, and the teacher has to count the present days by hand at the end of the semester. Students also do not know their attendance percentage until it is too late, and mistakes in records are common.

This thesis studies the problem of manual attendance and proposes a simple web-based student attendance management system. The scope of the work is a single department in a university, where teachers take attendance for their own courses and students check their own records.

Our approach is to design a system with three types of users: teacher, student and admin. The teacher marks attendance from a web page in a few clicks. The system saves the data in a database and calculates the attendance percentage of each student automatically. A student can log in and see the attendance of every course, and the admin can manage users and courses. The design follows a standard software development process of requirement collection, design, implementation and testing.

We expect that the system will save class time, reduce errors and give students a clear view of their own attendance. The results of the proposed system will be compared with the manual method in terms of time and accuracy. This report presents the introduction chapter, which explains the background, the problem, the objectives, the scope and the plan of the full thesis.

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Chapter I. Introduction

1.1 Background of the Study

Every university class starts with one small task: the teacher takes attendance. Attendance shows how regular a student is in class. In most universities, including those in Bangladesh, a student must attend a minimum percentage of classes to sit for the final exam. So attendance is not only a record. It has a direct effect on the result and the academic life of a student.

In many classrooms attendance is still taken on a paper sheet or in a notebook. The teacher calls the names one by one, or passes a sheet around the room, and later copies the data to a register. This method has worked for many years, but it has some clear weaknesses. It takes time from the lecture, it is easy to make mistakes, and the sheets can be damaged or lost.

Today almost every student and teacher uses a computer or a smartphone, and the internet is easy to access. Because of this, many daily tasks have moved online, such as shopping, banking and learning. Taking attendance can also be done in a simple online way. A web-based system keeps all the data in one place and can calculate results in a moment. This thesis is about the idea of building such a system for a university department.

1.2 Overview of the Current Situation

At present, most teachers follow one of two ways to record attendance. The first is the paper method, where names are called out or a sheet is signed by the students. The second is a spreadsheet, where the teacher types the data into a file on a personal computer after the class. Both ways are common, and both need extra work after the class.

Some institutions use special devices, such as fingerprint machines or card readers. These tools are fast, but they are costly to buy and to maintain. A small department may not have the budget for them. Some ready-made software also exists, but it is often made for big organizations and has many features that a single department does not need.

When we compare these options, we can see both good and bad sides. The paper method is cheap and easy to learn, but it is slow and not safe. The spreadsheet is better, but only the teacher can see it, so students cannot check their own record. The special devices are fast, but they need money and training. A simple, low-cost and easy to use system is still missing for many departments.

This gap is the reason for our work. We do not want to build a big and complex product. We want a small system that does the main job well: take attendance quickly, store it safely and show it clearly to the people who need it.

Table 1.1 Comparison of the manual method and the proposed system

Feature	**Manual (paper) method**	**Proposed web-based system**
Time to take attendance	Several minutes of class time	A few clicks, about one minute
Record keeping	Paper sheets that can be lost	Saved safely in a database
Percentage calculation	Done by hand at the end	Done automatically
Student access	Only by asking the teacher	Student can log in and check anytime
Chance of error	High	Low

Table 1.1 shows the main differences between the manual method and the proposed system.

1.3 Problem Statement

The main problem is that the manual way of taking attendance is slow, can be wrong and is not open to students. Because of this, the following problems are seen in a typical class:

- Class time is lost every day because names are called one by one.
- Paper records can be lost, torn or changed by mistake.
- Counting the attendance percentage by hand at the end of the semester takes a long time and may be wrong.
- Students cannot check their own attendance, so they find out about a shortage too late.

We want to solve these problems with a simple web-based system. The study is limited to one department of a university and to the daily class attendance of its students and teachers.

1.4 Importance of the Research

This topic is important for three groups of people. For teachers, the system saves time and removes the burden of keeping paper registers. For students, it gives a clear view of their attendance, so they can attend more carefully and avoid shortage problems. For the department, it keeps correct and safe records that can be used for reports at any time.

The topic is also a good example of how a small piece of software can improve a daily routine. It does not need costly devices. A normal computer or a smartphone with an internet connection is enough. For this reason, the idea can be used by other departments and even by schools and colleges.

1.5 Research Questions

This thesis tries to answer the following questions:

1. How can a web-based system make the process of taking attendance faster and simpler than the manual method?
2. How can the system calculate and show the attendance percentage correctly and automatically?
3. What are the main features that teachers, students and the admin need in an attendance system?

1.6 Aim and Objectives

The main aim of this thesis is to design and develop a simple and low-cost web-based attendance management system for a university department. To reach this aim, we set the following objectives:

- To study the problems of the manual attendance method.
- To collect the requirements of teachers, students and the admin.

- To design the system and its database in a clear way.
- To build a working version where the teacher can mark attendance and the student can view it.
- To compare the proposed system with the manual method by time and accuracy.

1.7 Scope and Definition of Terms

The scope of this work covers a single department. The system will have three types of users. A teacher can log in, select a course and mark the students as present or absent. A student can log in and see the attendance of each course along with the percentage. An admin can add or remove users and courses. The system will not cover fingerprint scanning, face recognition or payment features, and these can be considered in future work.

Some terms used in this report are explained below to make the reading easy.

- Attendance: the record of whether a student was present or absent in a class.
- Web-based system: a program that runs in a web browser, so the user does not need to install anything.
- Database: an organized place where the data of the system is stored safely.
- User: a person who uses the system, such as a teacher, a student or an admin.

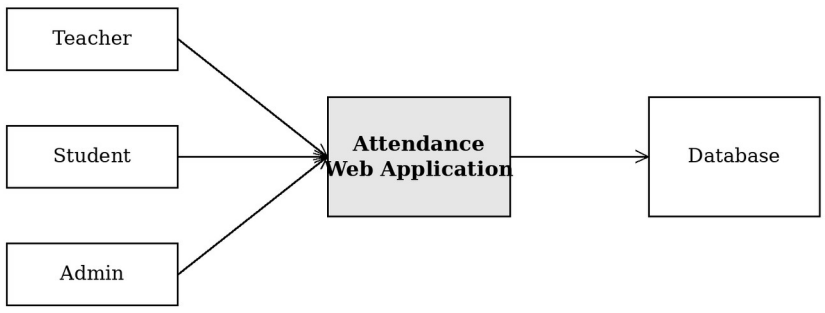


Figure 1.1 Overview of the proposed attendance system

Figure 1.1 shows that teachers, students and the admin use the same web application, and the application stores all data in one database.

1.8 Expected Outcomes

We expect the system to give three main benefits. The first is saving time. Instead of calling every name, the teacher will mark the students on a list, and the system will save the data at once. The second is better accuracy. Since the percentage is calculated by the system, there is no counting by hand, and fewer mistakes will happen. The third is more openness. Students will be able to see their own records, so they know where they stand in every course.

We also expect that the system will be easy to learn. A teacher who can use a normal website should be able to use it after a short introduction. The design will therefore keep the screens simple, with only the buttons that are really needed. We will check this idea during testing by asking a few teachers and students how easy the system was to use.

1.9 Limitations of the Study

Like every study, this work has some limits. First, the system is planned for one department only, so we cannot say how it will perform in a very large university with thousands of students. Second, the system depends on an internet connection. If the connection is down in a classroom, the teacher may not be able to mark attendance at that moment. Third, the system will trust the teacher to mark students correctly, and it will not check the identity of a student by any device.

These limits do not make the idea weak. They only show where the work stops. We have kept the scope small on purpose, so that we can finish the system, test it properly and explain every step clearly. The limits also point to useful future work, such as an offline mode and an automatic check of student identity.

1.10 Outline of the Methodology

We will follow a simple and well-known software development process. First, we will collect requirements by talking to teachers and students and by studying the current paper method. Second, we will design the system, including the screens and the database tables.

Third, we will build the system using common web tools. Fourth, we will test it with sample data to check that attendance is saved and the percentage is calculated correctly.

Finally, we will compare the results with the manual method. We will look at the time needed to take attendance and the number of mistakes. The full details of the method will be given in Chapter III.

1.11 Organization of the Thesis

This thesis is planned in five chapters. Chapter I is the introduction, which explains the background, the problem, the objectives and the scope. Chapter II is the literature review, which discusses earlier work on attendance systems and shows the gap that our work tries to fill. Chapter III explains the research methodology in detail. Chapter IV presents the results and discusses them. Chapter V concludes the thesis and gives some ideas for future work. The references are given at the end of the report.